

collision prevention, adaptive cruise control, and more, as options on their luxury sedan class of cars. As always, we can expect this technology to filter down to economy-class cars in the near future.

BMW, for instance, offers a “Remote Park Assist” feature: basically, you stop next to an available parking spot, get out, keep a button on the remote pressed, and watch as the car autoparks! If you feel the AI is being stupid, and your car is about to collide with another car or anything else, you just let go of the button and the car stops immediately!

A lot of manufacturers are also providing technology that tries and prevents *your car* from being tailgated. How does technology help prevent some other idiot from rear-ending you? Well it can't *stop* such accidents from happening, but it tries and prevents them. Sensors on the back of your car inform it of when another car is too close to you, and in such a case, your tail-lights start flashing and a warning light on your dashboard is set off. Different manufacturers have varying implementations: some actually compute the free space in front of you to slowly accelerate and increase the gap from the car behind; some just warn you, and set off a rear horn and flash tail-lights to inform the driver behind that he or she is too close for comfort.

Now before you start trying to imagine what havoc such technology will create in India, you



BMW's Hydrogen 7 series burns hydrogen in an internal combustion engine

should know that this feature is completely programmable—and it's not stupid! For instance, the acceptable distance between a speed of 0 and 30 kmph is obviously a lot less than what's acceptable at, say, 80 kmph. Plus, you have to imagine a day when every car has these systems: people would not be able to tailgate you even if they tried! Even at traffic signals, you would find nice and orderly, evenly-spaced-out car formations.

There are many different technologies that also help in minimising injuries when a crash is

inevitable: headrests are automatically adjusted when an impending crash is detected, to reduce whiplash; airbags are deployed with pressures that vary according to the speed you're going at, so as to prevent being hurt by the bag itself; seatbelts that change tension according to your speed, and many more!

The Future

Enough of what's already available in expensive cars today, what we really want to know is how close we are to the self-driving cars we've seen in movies and read about in tales of science fiction fantasy!

Surprisingly, we're not very far away.

Even in 2006, a lot of us cringe at the idea of letting a “dumb robotic car” drive us around at high speeds. But honestly, when you really think about it, don't computers already do that in the virtual world? Just pop in a racing simulator, or even an arcade racing game, and you'll find yourself playing against some very tough computer competition. They just seem to have a surreal amount of control on twisting and winding roads, and never seem to crash!

Yes, those are games, and it's different in real life where lives are at stake. Plus, as some will point out, the computer knows those roads very well, and is programmed to take a definite route, at a definite speed, following a pre-programmed racing line; but who said things have to be any different in the real world?

With GPS mapping being done of most cities of the world, and to an accuracy of a foot, what's to stop us from mapping it to, say, millimetre precision? You could conceivably feed all this GPS data to a car so it knows the routes and its position to the accuracy of a millimetre! Is that so unthinkable? It might take another 20 years, but it will eventually happen.

What about the changing nature of roads? Well, what about it? Honestly, cars today already have traction control, and know the difference between an icy surface, sand, tarmac, and rocky off-road adventures. But what if a kid runs across onto the road? Adaptive cruise control and collision prevention systems are already minimising the risk of accidents in such situations. Soon we should be able to improve it to the extent where it's better than human control. Much like braking, which is already computer-assisted in most new cars, humans are actually worse than computers at stopping suddenly. The automatic response of the human is to jam the brakes: a normal car would skid, lose control, and might even flip over on its side. It wouldn't stop in time either: studies have shown that a car takes longer to stop once it starts skidding. Computer-controlled braking “pumps” the brake several hundred times a second, preventing it from skidding and stopping the car faster and in a safer manner.

As for computer-controlled navigation in traffic, we have thus far only showcased *existing* technologies that are aiding in this. It's just a matter of refining them to the point where they're better and more reliable than humans. Sounds ridiculous, but driving is one aspect of